KOÇ UNIVERSITY

College of Engineering
COMP 491 Computer Engineering Design
Project Proposal

Domaino

COMP 491 - Computer Engineering Design

Project Proposal

Protein Sequence Based Domain Identification and Classification Platform

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Abstract

A protein is a chain of amino acids that folds into a three-dimensional object. Many proteins contain one or more domains, which are compact units that often fold independently and carry a recognizable structural, evolutionary, or functional role. Determining where each domain begins and ends is therefore a segmentation problem over the amino-acid sequence. Domaino will provide a complete sequence-based deep learning model and a software platform that predicts these boundaries and assigns each predicted domain a hierarchical CATH classification.

The final deliverable will include a deep-learning model training and evaluation pipeline, a user-inference service, and a web application. A user will submit an amino-acid sequence and receive a linear domain map, CATH labels, confidence scores, similar known domains, and evidence-linked functional suggestions. An optional structure-prediction stage will generate a three-dimensional visualization colored by the predicted domains. Performance will be measured on homology-aware held-out data and the CHAINSAW CATH1363 benchmark, with the design goal of matching or exceeding performance of the CHAINSAW model while requiring less input information.

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Section 1: Introduction

1.1 Concept

A protein is a large biological molecule that performs essential cellular tasks, including catalyzing reactions, transporting molecules, transmitting signals, and providing structural support. It is assembled as a linear chain of amino acids; once incorporated into the chain, each amino acid is called a residue. The residue order is encoded by genetic information and strongly influences how the protein chain folds into a three-dimensional structure and what biological functions it can perform. Computationally, a protein sequence can be represented as a string over an alphabet of twenty common amino-acid types.

Many proteins contain one or more domains. Domains are structural units that often retain recognizable roles across different proteins. Domains can be viewed as reusable modules whose combination and order contribute to the overall function of a protein. A domain may occupy one continuous sequence interval or several separated intervals that come together after folding. Determining which residues form each domain is therefore a segmentation problem. Each detected domain can then be classified using CATH, a four-level hierarchy classification of domains: Class, Architecture, Topology, and Homologous superfamily. CATH organizes domains according to increasingly specific structural and evolutionary similarities [1], [2].

Domaino aims to predict segmentation and classification together using only an amino-acid sequence of a newly discovered or designed protein. It will identify domain boundaries, group residues belonging to the same domain, and assign every detected domain a hierarchical CATH label. State-of-the-art segmentation systems such as CHAINSAW use a known or predicted structure, or a structure-derived distance matrix as input [3]. CATHe2, a recent state-of-the-art CATH classifier, assigns already isolated domains to known homologous superfamilies using sequence embeddings together with structure-derived information [4]. Domaino will aim to achieve competitive segmentation and classification performance when evaluated under the same benchmark data while doing both the segmentation and the classification task in a single model.

The proposed Domaino deep learning model will use a two-stage architecture. First, the pretrained ESM-2 protein language model will convert every residue into a numerical representation [5]. A segmentation network will estimate whether each residue belongs to a domain and whether pairs of residues belong to the same domain. A clustering procedure will transform these probabilities into complete domain assignments, including discontinuous domains. Second, the representations of all residues assigned to a domain will be combined and passed to a hierarchical classifier that predicts the four CATH levels in order.

For the training of our model, we will be using publicly available datasets: CATH, Pfam, and TED [1],[6],[7]. Experimentally derived CATH annotations will provide the primary domain boundaries and classification labels [1],[2]. Pfam regions will supply additional boundary supervision [6], while TED annotations derived from AlphaFold structures may be used as lower-weight pretraining data [7]. Homology-aware splitting will prevent closely related proteins from appearing in both training and evaluation sets. Segmentation will be evaluated against CHAINSAW using boundary, domain-count, and domain-overlap metrics. The CATH classification results will be compared with CATHe2 on the benchmark data used by the CATHe2 paper [4]. Performance of the model on long-sequence proteins and inference time will also be measured.

The expected outcome is a reproducible deep-learning and software platform. Through a web application, a scientist will be able to submit a sequence and receive the predicted domains, residue intervals, CATH labels, confidence scores, and similar annotated domains. Results will be displayed as a linear domain map and, optionally, on an independently predicted three-dimensional structure colored by domain.

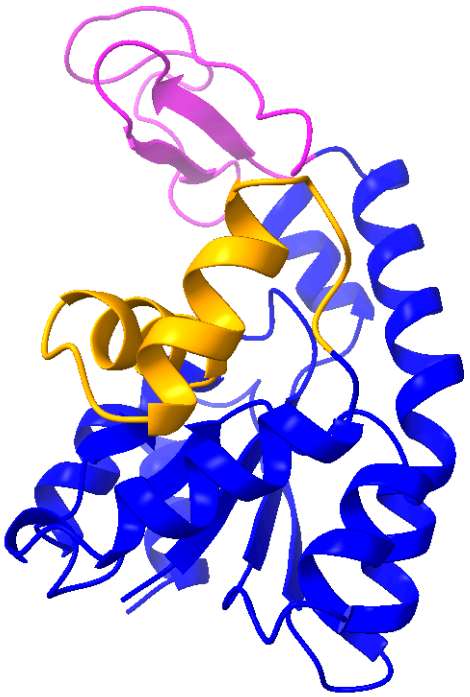


Figure 1: Three domains of a protein (adenylate kinase) highlighted in different colors:

the central catalytic CORE domain (blue; residues 1–29, 60–121, and 160–214),

the AMP-binding NMP domain (orange; residues 30–59),

and the ATP-site-covering LID domain (pink; residues 122–159). [2]

1.2 Objectives

The primary objective of this project is to design, implement, and evaluate Domaino as an integrated software platform that identifies and classifies protein domains directly from an amino-acid sequence. The intended users are molecular biologists, protein engineers, and other researchers who need domain information for newly discovered or designed proteins but may not have an experimentally determined or computationally predicted structure. Given a sequence, the completed system will identify the number and locations of its domains, group residues belonging to the same domain, assign hierarchical CATH labels, and report confidence and supporting database information.

By the end of Week 4, the project will produce a versioned data-processing pipeline that converts CATH, Pfam, and TED records into a common training format. Experimentally derived CATH annotations will be used as the main source of domain boundaries and classification labels. Pfam will provide additional boundary supervision, while TED data may be used for lower-weight pretraining. The pipeline will include automated validation, source tracking, and homology-aware train, validation, and test splits to reduce data leakage and enable reproducible experiments.

By Week 8, the project will produce a functional two-stage deep-learning model. The first stage will segment

the input sequence into one or more domains, including discontinuous domains where possible. The second stage will assign each predicted domain a consistent CATH classification at the Class, Architecture, Topology, and Homologous-superfamily levels. By Week 11, model selection and evaluation will be completed using held-out experimental CATH data and established benchmarks. Performance will be measured using domain overlap, boundary accuracy, domain-count accuracy, classification precision, recall and macro-F1 at each CATH level, confidence calibration, inference time, and performance on long or previously unseen proteins. The performance objective is to improve substantially upon the existing Domaino baseline, achieve segmentation results competitive with CHAINSAW, and approach CATHe2's classification performance under compatible evaluation conditions while requiring only sequence information.

By Week 12, the trained model will be integrated into a web-based inference platform. Users will be able to submit an amino-acid sequence and inspect a linear domain map, predicted boundaries, CATH labels, confidence scores, and similar annotated domains. The application will also provide an optional independently predicted three-dimensional structure in which the detected domains are highlighted. The structure will support interpretation and visualization but will not be used as input to the Domaino predictor.

By the final project demonstration in Week 14, the team will deliver the selected model checkpoint, reproducible training and evaluation pipeline, inference service, web interface, benchmark results, installation instructions, and user documentation. The project will be considered successful if the final system can process representative single-domain, multi-domain, and discontinuous-domain examples; return valid and interpretable results through the web interface; reproduce its reported evaluation results from versioned data and model configurations; match or exceed the performance of the state-of-the-art methods while being only sequence-based, and provide a documented comparison with CHAINSAW and CATHe2.

1.3 Background

Protein domains are structural and evolutionary units within proteins. Identifying these units is important because a domain often behaves as a reusable component with a particular molecular role or evolutionary origin. Domain annotations can help researchers understand an uncharacterized protein, compare it with known proteins, select regions for experimental study, and determine how a multidomain protein is organized. This requires solving two related problems: identifying which residues form each domain, known as domain segmentation, and assigning the detected domains to known structural and evolutionary groups, known as domain classification.

The scale of available protein-sequence data makes automated domain annotation increasingly important. The Encyclopedia of Domains processed approximately 214 million protein models from the AlphaFold Protein Structure Database and identified nearly 365 million predicted domains [7]. CATH organizes protein domains into a four-level hierarchy consisting of Class, Architecture, Topology, and Homologous superfamily, providing a standardized way to describe structural and evolutionary relationships [1], [2].

Current domain-segmentation methods predominantly use three-dimensional protein structure. Merizo represents a protein using its amino-acid sequence and three-dimensional residue coordinates, then applies invariant point attention to group residues into domains [8]. On its held-out CATH-663 benchmark, which contains 663 multidomain protein chains whose homologous superfamilies were excluded from its training data, Merizo obtained a domain-count mean absolute error of 0.332 and produced a decomposition matching either CATH or ECOD annotations for nearly 75% of proteins [8].

CHAINSAW similarly converts a protein structure into pairwise residue-distance and secondary-structure features and predicts whether each pair of residues belongs to the same domain [3]. It was evaluated on a benchmark of 1,365 experimentally determined protein chains from the Protein Data Bank [9], with CATH domain assignments used as ground truth. On this benchmark, CHAINSAW reported a mean intersection-over-union score of 0.88 over all proteins and 0.83 over multidomain proteins [3]. It also correctly parsed 78% of protein domains when a prediction was considered correct at a domain-level intersection over union of at least 0.8. The released CHAINSAW CATH benchmark will therefore be the main benchmark data for Domaino's segmentation component. Domaino will use the same measures, including mean intersection over union, correctly parsed domain percentage, domain-count error, and boundary-distance error.

For CATH classification, CATHe2 is a recent state-of-the-art method for assigning an already isolated protein domain to its CATH homologous superfamily [4]. CATHe2's highest-performing model combines embeddings from the amino-acid sequence with 3Di tokens, a structural alphabet derived from three-dimensional residue geometry. On a benchmark of approximately 1,700 CATH superfamilies, CATHe2 reported an accuracy of 92.2% and a macro F1-score of 82.3% [4]. CATHe2 therefore provides an important classification benchmark. However, it uses structural information and assumes that the correct domain boundaries have already been identified.

Domaino's classifier will be compared with CATHe2 using the corresponding CATH 4.3 domain sequences, labels, sequence-identity restrictions, and evaluation measures. It will first receive experimentally defined domain segments, isolating classification performance and permitting direct comparison using accuracy and macro F1-score. Domaino will then be evaluated end to end on complete protein chains, where it must locate and classify the domains. Reporting these evaluations separately will distinguish classification errors from segmentation errors.

There are sequence-based methods for domain-boundary prediction. DeepDom demonstrated that bidirectional recurrent networks can predict domain-boundary positions from sequence-derived features without using a three-dimensional structure [10]. More recently, Seq2DomML reported a mean domain intersection-over-union score of 0.846 on a homology-reduced bacterial dataset using residue-level sequence features and a bidirectional LSTM [11]. However, these methods address boundary prediction and do not simultaneously classify the detected domains within the complete CATH hierarchy. The unresolved problem is therefore whether our model can identify all of the domains of a given protein (including domains composed of multiple separated sequence intervals) and classify every detected domain within CATH by

only looking at the sequence information.

A direct sequence-based system is valuable because the amino-acid sequence is the most fundamental and widely available representation of a protein. Experimentally determining a protein structure is considerably more difficult than obtaining its sequence. Computational structure prediction with models such as AlphaFold has greatly increased structural coverage, but requiring an AlphaFold prediction introduces a separate inference pipeline, additional computation, and another source of uncertainty [12], [13]. Structure-prediction systems can also produce low-confidence regions, particularly for flexible or disordered sequences. In multidomain proteins, the structures of the individual domains may be predicted accurately while their relative positions and orientations remain uncertain [13]. A structure-based domain predictor may consequently inherit errors or uncertainties from the upstream structure model.

A sequence-only model would allow large collections of sequences to be screened directly and would provide evidence that is independent of an AlphaFold prediction. Predicted structures could still be generated afterward for visualization or comparison, but they would not influence Domaino's primary segmentation and classification results. This distinction is particularly useful when structure prediction is computationally inconvenient, when an existing structure has low-confidence regions, or when researchers want to compare independent sequence-based and structure-based evidence.

The intended users include bioinformaticians performing large-scale sequence analysis, structural biologists selecting domain constructs for experiments, molecular biologists investigating proteins with limited annotation, and protein engineers evaluating newly designed sequences. Domaino's predictions could help these users identify likely domain boundaries, detect similarities to characterized CATH groups, select experimentally relevant regions, and develop hypotheses about possible functions or evolutionary origins. A CATH classification or similarity to a known domain alone would not be enough to prove that the protein performs the same function. However, it would provide an interpretable source of evidence that researchers could combine with sequence similarity, structural predictions, literature, and experimental results.

The project is timely because generative protein-design methods can now produce large numbers of previously unseen protein candidates. ProteinMPNN generates amino-acid sequences intended to fold into specified protein backbones [14], while RFdiffusion generates new protein structures and functional scaffolds for applications such as molecular binding, oligomer formation, and motif presentation [15]. These systems allow researchers to create candidate proteins more rapidly than those candidates can be experimentally produced and validated. Furthermore, a computationally designed protein may have no direct natural counterpart and may therefore lack the database annotations normally transferred from known relatives.

The designed proteins could be submitted to Domaino before synthesis and experimental validation. Domaino could indicate whether a sequence contains coherent domain-like regions, whether those regions resemble known CATH groups, and what biological roles those similarities might suggest. This would provide an indirect interpretation of a designed protein's possible organization and function. Importantly, this analysis would not require every candidate first to be processed by AlphaFold or another structure-prediction model; avoiding another source of uncertainty and other problems previously mentioned, such as low-confidence regions produced by structure-prediction models.

Section 2: Scientific and Technical Methodology and Associated Work Plan

2.1 Methodology

Relevance and overall approach. Domaino will be developed in the following stages: data preparation, baseline implementation, model design, controlled training, evaluation, software integration, and final validation. Each stage will produce testable outputs before the next stage begins, reducing the risk of

discovering data or integration problems later onto the project.

The project is relevant because amino-acid sequences are generally available before experimentally determined or computationally predicted structures. Therefore, a sequence-only model could support earlier and larger-scale analysis of newly discovered and computationally designed proteins without requiring AlphaFold or another structure-prediction model as a prerequisite. Its outputs could help researchers identify domain organization, select regions for experiments, and form evidence-based hypotheses about structural, functional, or evolutionary relationships.

Data preparation and supervision. The exact releases of CATH, Pfam, TED, and related sequence resources will first be downloaded, versioned, and converted into a common representation. PDB-derived CATH domains will provide the primary supervision because they contain experimentally supported structural boundaries and hierarchical CATH labels [1], [2]. Pfam regions will be used only as auxiliary sequence-region evidence because they do not provide CATH classifications and do not always correspond exactly to structural domains [6]. TED annotations originate from AlphaFold-predicted structures and will therefore be considered only for lower-weight pretraining if they improve later performance on experimental CATH data [7].

Each record will retain its source, and source-specific loss masks will prevent missing labels from being treated as negative examples. Automated checks will detect invalid sequences, duplicate records, inconsistent coordinates, overlapping annotations, and missing labels. This approach will allow the system to benefit from multiple data sources.

To prevent overestimated performance, sequences will not be divided randomly. CATH sequence clusters or MMseqs2 similarity clusters will keep close homologues within the same partition [16]. The CHAINSAW segmentation benchmark and CATHe2 classification test domains, together with their close homologues, will be excluded before training. Training, validation, and test manifests will be versioned so that experiments can be reconstructed.

Sequence representation and baseline models. Simple baselines will be implemented before the full architecture. These will include treating the complete sequence as one domain, a residue-level boundary predictor, and a classifier that predicts CATH labels from experimentally defined domain segments. They will provide minimum reference results against which the effectiveness of later design choices can be measured.

The full system will use a pretrained ESM-2 protein language model to convert every amino-acid residue into a contextual numerical representation [5]. ESM-2 is suitable because it learns informative sequence patterns from large protein collections without requiring three-dimensional structure. The model size will be selected according to available memory, inference time, and validation performance. The ESM-2 encoder will initially remain frozen to reduce training cost; limited fine-tuning will be considered only if it produces a clear validation improvement.

Domain segmentation. Two candidate segmentation architectures will be considered. The first, lower-risk design will predict both residue-level and residue-pair information. The residue output will estimate whether each residue belongs to a domain, a boundary, or an unassigned linker region. The pairwise output will estimate whether two residues belong to the same domain. Pairwise prediction is included because direct boundary prediction alone has difficulty representing discontinuous domains, whose residues occur in separated sequence intervals. It is also independent of the arbitrary numbering assigned to individual domains. For the pairwise design, a clustering procedure will convert the predicted probabilities into complete domain assignments. It will apply constraints such as minimum domain length, residue coverage, acceptable linker gaps, and consistency among predicted intervals. Since the pairwise representation grows quadratically with sequence length, the implementation will use length-based batching, mixed-precision training, pair sampling, and overlapping windows where necessary.

A second candidate will adapt a set-prediction approach used by DETR and MaskFormer for computer-vision tasks [17], [18]; to one-dimensional protein sequences. A fixed set of learned domain queries will attend to the ESM-2 residue embeddings. Each query will predict a residue-membership mask and either a domain-confidence or "no domain" score. Because a mask can select separated sequence positions, this design can also represent discontinuous domains. Bipartite matching will pair the unordered predicted masks with the ground-truth domains during training. This approach avoids producing an explicit residue-pair matrix and may remove the need for a separate clustering procedure, but it introduces a maximum number of domain queries and additional training complexity.

The two architectures will first be compared on the same CATH development subset using segmentation accuracy and discontinuous-domain performance. The architecture selected for full-scale training will be the one that provides the best predictive performance. The maximum supported sequence length will be measured experimentally.

Hierarchical CATH classification. After segmentation, the residue embeddings assigned to each predicted domain will be combined into a domain-level representation. If the query-based architecture is selected, its learned domain-query representation may also be included. A classification network will predict the Class, Architecture, and Topology levels as the primary outputs. Separate outputs will be connected through a hierarchy-consistency rule so that a detailed prediction cannot contradict its broader parent classes.

The Homologous-superfamily level (H) will be treated as a conditional, finer-grained prediction. It is the most specific CATH level and contains many rare or poorly represented classes. An H-level label will therefore be returned only when the predicted superfamily is represented adequately during training and its calibrated confidence exceeds a selected threshold. Otherwise, the model will retain its C, A, and T predictions while reporting the H level as unresolved.

Class-balanced or focal loss will be evaluated because the CATH dataset is highly imbalanced: most homologous superfamilies contain relatively few annotated domains, while a small number of highly populated superfamilies contain a large proportion of the training examples. Source and label masks will ensure that unsupported or unavailable H-level labels are not treated as negative examples.

The classifier will be evaluated in two modes. First, it will receive experimentally defined domain segments, isolating classification performance and allowing direct comparison with CATHe2. Second, it will receive Domaino's predicted segments, measuring the complete sequence-to-domain-to-CATH workflow. For the CATHe2-compatible benchmark, forced top-one H-level accuracy and macro F1-score will be reported. For confidence-filtered H predictions, coverage (the proportion of domains receiving an H label) will be reported alongside accuracy so that abstaining from uncertain predictions cannot artificially inflate performance. Reporting these evaluation modes separately will show whether an error originates from segmentation, classification, or H-level uncertainty.

Training and model selection. Training will begin with CATH-only baselines and small-scale versions of the pairwise and query-based segmentation architectures. Both candidates will use the same ESM-2 representations, dataset partitions, and evaluation measures. Their segmentation performance, CATH classification performance, discontinuous-domain behavior will determine which architecture proceeds to full-scale training.

After the architecture is selected, Pfam auxiliary supervision and optional TED pretraining will be introduced as controlled experiments. The contributions of the selected segmentation design, hierarchical classification, class balancing, source masks, confidence-based H-level abstention, and post-processing rules will be measured through ablation experiments in which one component is changed at a time. Random seeds, dataset manifests, hyperparameters, learning curves, checkpoints, and failed experiments will be recorded.

Model parameters and post-processing thresholds will be selected using validation data only. Once the final configuration has been selected, its architecture, checkpoint, preprocessing rules, segmentation thresholds,

and classification-confidence thresholds will be frozen.

Performance evaluation. Segmentation will be evaluated on held-out experimental CATH data and the published CHAINSAW benchmark while providing Domaino only with amino-acid sequences [3]. Measures will include mean intersection over union, the percentage of correctly parsed domains at an intersection-over-union threshold of 0.8, boundary-position error, and domain-count error.

Classification will be evaluated on the CATHe2-compatible domain benchmark using accuracy and macro F1-score at each CATH level [4]. Results will be reported separately for single-domain, multidomain, continuous-domain, discontinuous-domain, long, and low-similarity proteins. End-to-end performance, confidence calibration, inference time, peak memory use, and CPU and GPU behavior will also be measured.

Software platform and system validation. The selected model will be exposed through an inference service and a web interface. The backend will validate sequences, execute inference, store structured results, and return clear error states. CPU inference will be supported for individual proteins, while GPU acceleration may be used for training, longer sequences, or concurrent requests. The interface will display a linear domain map, CATH labels, confidence values, similar known domains, and evidence-linked functional suggestions.

Optional structure prediction will run separately, only for visualization purposes and will not provide input to Domaino's segmentation or classification model. When a predicted structure is available, its residues will be colored using Domaino's domain assignments. Separating this stage ensures that structure-prediction delays or failures do not prevent users from receiving the primary results.

Effectiveness and novelty. The methodology is designed to be effective through complementary engineering choices. ESM-2 embeddings will provide residue-level sequence context, pairwise prediction will support multidomain and discontinuous-domain segmentation, and hierarchical classification will prevent incompatible CATH labels. Experimental CATH data will provide the principal supervision, while source masks will allow Pfam and TED to be used without confusing their different annotation types. Homology-aware splits, baseline comparisons, ablation experiments, confidence calibration, and matched evaluation against CHAINSAW and CATHe2 will determine whether each design choice provides a measurable benefit.

The novelty comes from combining sequence-only segmentation, discontinuous-domain handling, and hierarchical CATH classification in one end-to-end model and deployable software platform. Existing high-performing systems generally use structural information, address only segmentation, or assume that domain segments are already known. Domaino will instead begin with a complete raw sequence and produce both domain assignments and CATH labels, while using predicted structure only afterward as an optional visualization.

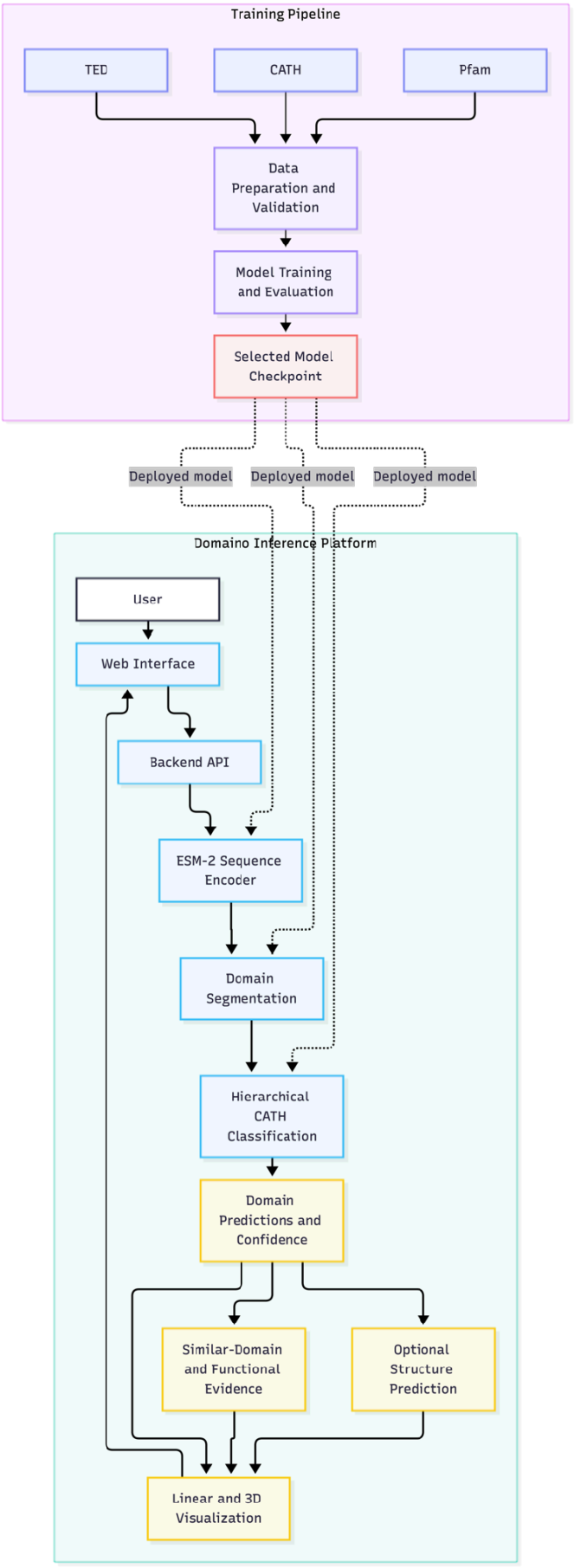


Figure 2: Planned System Architecture of the Domaino ML Model and Web Application

Table 1: Planned Design Documents, Scope, Tools, and Preparation Schedule

#	Design Document	Content / Scope	Format / Tool	Needed By (Week)
1	Context diagram and Data Flow Diagram	CATH/Pfam/TED → parsers → common schema → audit → split manifests → training → checkpoint → inference	Mermaid (flowchart)	4
2	Dataset ER diagram and schema description	Sequence, domain intervals, source, CATH labels, split assignment, source masks, discontinuous-domain encoding	Mermaid (erDiagram) + markdown tables	4
3	Use case diagram and actor descriptions	Actors: molecular biologist, protein engineer, bioinformatician, maintainer; main and alternative flows	Mermaid / draw.io	5
4	System architecture, component and deployment diagrams	Inference worker, backend, frontend, optional structure service, local data store; containerized deployment	draw.io (Mermaid if simple)	6
5	Model architecture diagram and ML class diagram	ESM-2 embedder, segmentation network, clusterer, hierarchical classifier, losses and source masks	Mermaid (classDiagram) + draw.io	6
6	Sequence diagrams	Submit → job → inference → result; optional structure job; failure and fallback paths (Risks 6 and 7)	Mermaid (sequenceDiagram)	7
7	UI wireframes and user-flow diagram	Input, validation error, progress, linear domain map, similar-domain panel, 3D view	Figma	7
8	Application ER diagram and database table descriptions	Jobs, submitted sequences, predictions, domains, CATH labels + confidence, structure jobs, cached similarity results, version/provenance records, exports	Mermaid (erDiagram) + DBML/markdown per table	7
9	API contract and state diagrams	OpenAPI / JSON schema; job and structure-job states (queued, running, completed, failed)	OpenAPI YAML + Mermaid (stateDiagram)	7

2.2 Work Package Descriptions and Task Distributions

Participant numbers are P1 Eylül Sevinç Sevinç, P2 Umut Altınkaynak, and P3 Erkmen Erken. All three participants contribute to every package.

Work Package 1: Data Integration and Quality

Work package number	**1**	Start date or starting event:	**Week 4**
Work package title	Data Integration and Quality		
Participant number	1	2	3
Participant name	Eylül Sevinç Sevinç	Erkmen Erken	Umut Altınkaynak
Weeks per participant	1	1	1

Objectives

- Obtain reproducible versions of the CATH, Pfam, TED, and related protein-sequence data.
- Transform the different data sources into a common representation.
- Prevent risks associated with training data listed under *2.5.Risk Analysis*: mitigate label leakage and source confusion.
- Construct homology-aware training, validation, and test partitions that support fair comparison with CHAINSAW and CATHe2.

Description of work

T1.1 **(W4–W14) Establish and maintain the project-management and development infrastructure.**

The team will establish the code repository, Notion workspace, development process, and project web page. Tasks, decisions, risks, progress, and meetings with the course staff and project advisor will be documented throughout the semester.

T1.2 (**W4) Select, download, and version the biological data sources.**

The exact releases of CATH, Pfam, TED, PDB, and related resources will be downloaded with their source, version, checksum, and license information. CATH will provide the main ground truth, Pfam will provide auxiliary supervision, and TED will be treated as lower-confidence data derived from predicted structures.

T1.3 (W4) Design and implement a common dataset schema.

Parsers will convert each source into a shared format containing the sequence, domain intervals, source information, and available CATH labels. Residue coordinates will be standardized, and the schema will support single, multiple, and discontinuous domains.

T1.4 (W4) Audit and clean the converted data.

Checks will be performed to detect invalid sequences, duplicate records, missing labels, incorrect boundaries, and inconsistent lengths. Conflicts between CATH, Pfam, and TED will be measured and

documented, and representative records will be inspected manually.

T1.5 (W4) Construct leakage-safe training, validation, and test partitions.

Proteins will be grouped using CATH or MMseqs2 similarity clusters so that close homologues remain in the same partition. CHAINSAW and CATHe2 benchmark examples and their close homologues will be excluded from training, and versioned dataset manifests will be generated.

Deliverables

D1.1 (W4): Versioned data-download and conversion pipeline with source records, checksums, and reproducible configuration files.

D1.2 (W4): Dataset manifests, data card, summary statistics, annotation-conflict report, and homology-leakage audit.

Milestones

M1.1 (Middle of W4): The repository and Notion workspace are established.

M1.2 (End of W4): Representative CATH, Pfam, and TED records pass coordinate-conversion and schema-validation tests.

M1.3 (End of W4): Dataset version 1 and its training, validation, test, and benchmark manifests are frozen for model development.

Table 2: Task Distribution Matrix for Work Package 1

Task ID	Task	Scheduled Weeks	Coordinating Lead	Other Contributors	Rationale / Shared-work arrangement
T1.1	Establish and maintain project-management and development infrastructure	W4–W14	P1	P2, P3	Shared coordination; P1 maintains project records and infrastructure.
T1.2	Select, download, and version biological data sources	W4	P2	P1, P3	P2 coordinates data-source/version tracking; all verify source metadata.
T1.3	Design and implement a common dataset schema	W4	P3	P1, P2	P3 coordinates schema implementation; all review representation and coordinates.

T1.4	Audit and clean converted data	W4	P1	P2, P3	P1 coordinates quality checks; all inspect conflicts and representative records.
T1.5	Construct leakage-safe training, validation, and test partitions	W4	P2	P1, P3	P2 coordinates partitioning; all verify benchmark exclusions and manifests.

Work Package 2: Baseline and Model Architecture

Work package number	**2**	Start date or starting event:	**Week 5**
Work package title	Baseline and Model Architecture		
Participant number	1	2	3
Participant name	Eylül Sevinç Sevinç	Erkmen Erken	Umut Altınkaynak
Weeks per participant	3	3	3

Objectives

- Implement simple segmentation and classification baselines from scratch to establish minimum reference performance.
- Design a sequence-only architecture that performs both domain segmentation and hierarchical CATH classification.
- Define target representations and loss functions that support proteins of different lengths and domains composed of separated sequence intervals.
- Ensure that the proposed model can be trained within the available GPU-memory and semester-time constraints.
- Produce a tested model implementation that is ready for full-scale experiments.

Description of work

T2.1 (W5) Implement minimum reference baselines.

Simple segmentation and classification models will be implemented to establish minimum reference scores. These will include a one-domain assumption, a residue-level boundary predictor, and a domain classifier based on pooled sequence embeddings.

T2.2 (W5) Implement the protein-sequence representation pipeline.

A suitable ESM-2 model will generate one embedding for each residue. The pipeline will handle tokenization, padding, masking, batching, and optional embedding caching.

T2.3 (W5–W6) Design and implement the domain-segmentation network.

The network will predict residue assignments and pairwise same-domain probabilities. The pairwise

representation will support multidomain and discontinuous-domain predictions.

T2.4 (W6) Implement conversion from network outputs to domain assignments.

A clustering component will convert model probabilities into domain intervals. It will apply minimum-length, linker, residue-coverage, and consistency constraints.

T2.5 (W6–W7) Design and implement the hierarchical CATH classifier.

Domain embeddings will be classified at the Class, Architecture, Topology, and Homologous-superfamily levels. Hierarchical consistency and unknown outputs will prevent unsupported classifications.

T2.6 (W7) Implement source-aware training targets and loss functions.

Separate losses will be implemented for residue assignment, boundaries, pairwise grouping, and CATH classification. Source masks will control which losses apply to CATH, Pfam, and TED records.

T2.7 (W7) Implement memory controls and model-level testing.

Length-based batching, mixed precision, pair sampling, overlapping windows, and gradient accumulation will be tested. Automated tests will cover masks, losses, clustering, checkpoint handling, and inference.

Deliverables

D2.1 (W5): Implemented segmentation and CATH-classification baselines with a baseline-results report.

D2.2 (W6): Model-architecture specification describing the residue representation, pairwise segmentation, clustering, classification hierarchy, losses, and uncertainty handling.

D2.3 (W7): Tested model version 1, training configuration, checkpoint format, inference-output schema, and automated model test suite.

Milestones

M2.1 (End of W5): Both baseline systems train and produce valid predictions on a small data subset.

M2.2 (End of W6): The segmentation, clustering, and classification components operate together on controlled examples.

M2.3 (End of W7): The complete architecture fits a small controlled dataset and produces valid assignments without exceeding the declared memory limit.

Table 3: Task Distribution Matrix for Work Package 2

Task ID	Task	Scheduled Weeks	Coordinating Lead	Other Contributors	Rationale / Shared-work arrangement
T2.1	Implement minimum reference baselines	W5	P3	P1, P2	P3 coordinates baseline implementation and result recording; all contribute and review results.
T2.2	Implement protein-sequence representation pipeline	W5	P2	P1, P3	P2 coordinates ESM-2 embedding pipeline; all test sample inputs.
T2.3	Design and implement domain-segmentation network	W5–W6	P3	P1, P2	P3 coordinates segmentation implementation; all review output behavior.
T2.4	Convert network outputs to domain assignments	W6	P1	P2, P3	P1 coordinates clustering/post-processing; all validate controlled examples.
T2.5	Design and implement hierarchical CATH classifier	W6–W7	P2	P1, P3	P2 coordinates classifier; all review hierarchy and unknown outputs.
T2.6	Implement source-aware training targets and loss functions	W7	P3	P1, P2	P3 coordinates loss/mask implementation; all verify source-specific behavior.
T2.7	Implement memory controls and model-level testing	W7	P1	P2, P3	P1 coordinates testing/memory checks; all run and review tests.

Work Package 3: Training, Evaluation, and Model Selection

Work package number	**2**	Start date or starting event:	**Week 8**
Work package title	Training, Evaluation, and Model Selection		
Participant number	1	2	3
Participant name	Eylül Sevinç Sevinç	Erkmen Erken	Umut Altınkaynak
Weeks per participant	3	5	3

Objectives

- Train the Domaino models from scratch using reproducible experiment configurations.
- Determine whether auxiliary Pfam supervision or optional TED pretraining improves performance over CATH-only training.
- Measure the contribution of the pairwise representation, hierarchical classification, class balancing, and clustering components.
- Select one final model using validation data without repeatedly inspecting the external test benchmarks.
- Evaluate the final model against the CHAINSAW segmentation and CATHe2 classification benchmarks.

Description of work

T3.1 (W8) Define the controlled training and experiment protocol.

The team will fix the random seeds, dataset versions, training rules, computing budget, and checkpoint-selection criteria. Each run will record its configuration, code revision, runtime, and result.

T3.2 (W8–W9) Train the CATH-only reference and full models.

The baseline and complete models will be trained using CATH data under the same conditions. Results will be separated by domain count and domain type.

T3.3 (W9–W10) Evaluate alternative uses of Pfam and TED.

Pfam will be tested as masked auxiliary supervision. If time and computing resources permit, TED pretraining followed by CATH fine-tuning will also be evaluated.

T3.4 (W9–W10) Conduct architectural and loss-function ablations.

Experiments will measure the contributions of pairwise prediction, hierarchical classification, class balancing, source masks, and clustering choices.

T3.5 (W10) Tune post-processing thresholds and calibrate confidence.

Boundary, clustering, minimum-length, and confidence thresholds will be selected using validation data only.

T3.6 (W11) Select and freeze the final model.

The final model will be selected using validation performance, multidomain accuracy, computational cost, and stability. Its weights, architecture, thresholds, and preprocessing configuration will then be frozen.

T3.7 (W11–W12) Perform final benchmark evaluation.

Segmentation will be evaluated on the CHAINSAW CATH benchmark, and classification will be evaluated on the CATHe2 test domains. A separate evaluation will measure complete end-to-end performance.

Deliverables

D3.1 (W10): Experiment register and ablation report containing configurations, learning curves, validation metrics, computational costs, and failed-run explanations.

D3.2 (W11): Frozen final checkpoint, preprocessing and inference configuration, model card, confidence thresholds, and reproducibility record.

D3.3 (W12): Final benchmark report containing CHAINSAW-compatible segmentation results, CATHe2-compatible classification results, end-to-end results, and limitations.

Milestones

M3.1 (End of W9): At least one complete CATH training run finishes reproducibly without data, loss-mask, or checkpoint errors.

M3.2 (End of W10): The required data-source and architecture comparisons are completed.

M3.3 (End of W11): The final model and all decision thresholds are frozen before benchmark testing.

M3.4 (End of W12): External benchmark evaluation is complete and reproducible from the frozen checkpoint.

Table 4: Task Distribution Matrix for Work Package 3

Task ID	Task	Scheduled Weeks	Coordinating Lead	Other Contributors	Rationale / Shared-work arrangement
T3.1	Define controlled training and experiment protocol	W8	P3	P1, P2	P3 coordinates experiment protocol/configuration; all agree on protocol.
T3.2	Train CATH-only reference and full models	W8–W9	P1	P2, P3	P1 coordinates reproducible training runs; all monitor and interpret results.
T3.3	Evaluate alternative uses of Pfam and TED	W9–W10	P2	P1, P3	P2 coordinates data-source comparisons; all review feasibility and findings.
T3.4	Conduct architectural and loss-functio n ablations	W9–W10	P2	P1, P3	P2 coordinates ablation runs, consistent with P2's larger planned WP3 workload (5 weeks vs 3 for P1/P3).
T3.5	Tune post-process ing thresholds and calibrate confidence	W10	P3	P1, P2	P3 coordinates validation-only threshold selection; all check for test-set leakage.
T3.6	Select and freeze final model	W11	P2	P1, P3	P2 coordinates selection and freeze; all review validation evidence.

T3.7	Perform final benchmark evaluation	W11–W12	P1	P2, P3	P1 coordinates benchmark execution/reporting; all verify reproducibility and limitations.

Work Package 4: Web Platform and Visualization

Work package number	**2**	Start date or starting event:	**Week 7**
Work package title	Web Platform and Visualization		
Participant number	1	2	3
Participant name	Eylül Sevinç Sevinç	Erkmen Erken	Umut Altınkaynak
Weeks per participant	5	5	5

Objectives

- Convert the sequence-based models into an accessible end-to-end software platform.
- Allow users to submit a protein sequence and inspect domain boundaries, CATH labels, confidence values, and supporting evidence.
- Present the same domain assignments consistently in a linear sequence view and an optional three-dimensional structure view.
- Keep the core sequence-only prediction available even when optional structure prediction or external information services fail.
- Provide clear error handling, provenance, privacy information, and exportable results.

Description of work

T4.1 (W7) Design the user workflow, interface wireframes, and API contract.

The workflow from sequence submission to result interpretation will be designed. Wireframes and a JSON schema will define communication among the model, backend, and frontend.

T4.2 (W7–W8) Implement the backend inference service.

The backend will validate input, create prediction jobs, invoke the model, store results, and report job status. Clear errors will be returned when a request cannot be completed.

T4.3 (W8–W9) Implement the core frontend.

The frontend will provide sequence input, FASTA upload, validation feedback, progress information, and result pages containing domain maps, boundaries, labels, and confidence scores.

T4.4 (W9–W10) Implement domain-similarity and functional-evidence presentation.

Predicted domains will be compared with known CATH domains. Similar domains, supporting annotations, similarity values, and source links will be displayed.

T4.5 (W10–W11) Implement optional structure prediction and three-dimensional visualization.

Structure prediction will run separately from core inference. Completed structures will be displayed using the same domain colors as the linear map.

T4.6 (W11–W12) Add deployment, privacy, provenance, and export functions.

The application will record relevant versions, protect submitted sequences, provide result exports, and include a containerized deployment configuration.

T4.7 (W12–W13) Test usability and system behavior.

The team will test invalid and long inputs, model failures, unavailable services, browser compatibility, loading states, and visualization consistency.

Deliverables

D4.1 (W7): User-flow specification, interface wireframes, versioned API schema, and interactive interface using mocked predictions.

D4.2 (W11): Core web application in which a submitted sequence reaches the model and returns a domain map, CATH labels, and confidence values.

D4.3 (W13): Deployable end-to-end web platform with similarity information, optional three-dimensional visualization, exports, provenance, privacy controls, and installation documentation.

Milestones

M4.1 (End of W7): A user can submit a sequence to the mocked system and inspect a simulated result.

M4.2 (End of W11): The web application invokes the selected model and displays its segmentation and classification output correctly.

M4.3 (End of W12): Domain residue mappings and colors agree between the sequence and three-dimensional views.

M4.4 (End of W13): The application can be installed in a clean environment and complete the intended

demonstration workflow.

Table 5: Task Distribution Matrix for Work Package 4

Task ID	Task	Scheduled Weeks	Coordinating Lead	Other Contributors	Rationale / Shared-work arrangement
T4.1	Design user workflow, interface wireframes, and API contract	W7	P3	P1, P2	P3 coordinates UI/API design; all agree on end-to-end workflow.
T4.2	Implement backend inference service	W7–W8	P2	P1, P3	P2 coordinates backend/model interface; all test valid and invalid requests.
T4.3	Implement core frontend	W8–W9	P1	P2, P3	P1 coordinates frontend implementation; all review usability and outputs.
T4.4	Implement domain-similarity and functional-evidence presentation	W9–W10	P3	P1, P2	P3 coordinates similarity/evidence display; all verify sources and values.
T4.5	Implement optional structure prediction and 3D visualization	W10–W11	P2	P1, P3	P2 coordinates integration of optional 3D functionality; all check matching domain colors.
T4.6	Add deployment, privacy, provenance, and export functions	W11–W12	P1	P2, P3	P1 coordinates release/privacy/export checklist; all test the workflow.
T4.7	Test usability and system behavior	W12–W13	P3	P1, P2	P3 coordinates test cases; all run and document issues.

Work Package 5: Integration, Demonstration, and Documentation

Work package number	**2**	Start date or starting event:	**Week 12**
Work package title	Integration, Demonstration, and Documentation		
Participant number	1	2	3
Participant name	Eylül Sevinç Sevinç	Erkmen Erken	Umut Altınkaynak
Weeks per participant	3	3	3

Objectives

- Validate the complete system under conditions representative of the final demonstration.
- Resolve integration failures among data preprocessing, model inference, backend services, frontend visualization, and optional structure prediction.
- Prepare a clear demonstration showing both the engineering design and the scientific usefulness of the system.
- Complete the required project web page, repository, demo video, poster, presentation, technical documentation, and final report.
- Use the additional finals week to address feedback and submit a reproducible final implementation.

Description of work

T5.1 (W12) Conduct end-to-end integration and failure testing.

The complete workflow will be tested using valid, invalid, short, long, single-domain, multidomain, and discontinuous-domain inputs. Integration failures will be recorded and prioritized.

T5.2 (W12–W13) Prepare the demonstration dataset and expected outputs.

Representative proteins will be selected for the main prediction cases, including a designed protein. Expected outputs and fallback examples will be prepared.

T5.3 (W13) Verify reproducibility and deployment.

A clean installation will be completed using the written instructions. Model files, configurations, software versions, and hardware requirements will be verified.

T5.4 (W13–W14) Prepare user, technical, and course documentation.

The team will complete the user guide, technical documentation, project web page, repository documentation, and demonstration video.

T5.5 (W13–W14) Prepare the poster and final presentation.

The poster and presentation will summarize the problem, design, datasets, model, evaluation, interface, results, and limitations. At least two complete rehearsals will be conducted.

T5.6 (W14) Freeze and present the demonstration release.

T5.7 (W15 – finals week) Complete post-demonstration corrections and final submission.

The team will address feedback, correct remaining problems, rerun the tests, complete the final report, and submit the final implementation.

Deliverables

D5.1 (W13): Tested release candidate, integration-test report, clean-install record, demonstration dataset, expected-output document, and fallback materials.

D5.2 (W14): Demonstration release, completed project web page, documented code repository, demo video, poster, presentation, user guide, and archived benchmark results.

D5.3 (W15): Final corrected software release, final report, complete implementation submission, technical documentation, and reproducibility archive.

Milestones

M5.1 (End of W12): The complete system passes the defined core integration tests.

M5.3 (W14): The final demonstration and poster presentation are completed.

M5.4 (W15): Demonstration feedback is addressed, all release tests pass, and the final report and implementation package are submitted.

Table 6: Task Distribution Matrix for Work Package 5

Task ID	Task	Scheduled Weeks	Coordinating Lead	Other Contributors	Rationale / Shared-work arrangement
T5.1	Conduct end-to-end integration and failure testing	W12	P1	P2, P3	P1 coordinates integration test log; all test the end-to-end system.
T5.2	Prepare demonstration dataset and expected outputs	W12–W13	P2	P1, P3	P2 coordinates selection of examples; all verify expected/fallback outputs.
T5.3	Verify reproducibility and deployment	W13	P3	P1, P2	P3 coordinates clean-install check; all verify instructions and dependencies.
T5.4	Prepare user, technical, and course documentation	W13–W14	P1	P2, P3	P1 coordinates documentation checklist; all contribute/review their sections.
T5.5	Prepare poster and final presentation	W13–W14	P2	P1, P3	P2 coordinates poster/presentation assembly; all provide content and rehearse.
T5.6	Freeze and present demonstration release	W14	P3	P1, P2	P3 coordinates release checklist; all participate in demonstration.
T5.7	Complete post-demonstration corrections and final submission	W15	P1	P2, P3	P1 coordinates final checklist; all address feedback and verify submission.

2.3 Demonstration

Performance analysis will be conducted on fixed held-out datasets after the final model and decision thresholds have been frozen. Segmentation performance will be measured using mean intersection over union, the percentage of correctly identified domains at an intersection-over-union threshold of 0.8, boundary-position error, and domain-count error. Classification will be measured using accuracy and macro F1-score at each CATH level. An end-to-end measure will report how often a domain is both located correctly and assigned the correct CATH label. Results will be reported separately for single-domain,

multidomain, and discontinuous-domain proteins. Inference time, peak memory use, supported sequence length, and confidence calibration will also be measured. Domaino will be compared with CHAINSAW for segmentation and CATHe2 for classification using their corresponding benchmark datasets.

The final demonstration will show both model inference and software integration. We will prepare a small fixed set containing an experimentally annotated single-domain protein, a multidomain protein, a protein with a discontinuous domain if available, and a designed or recently characterized sequence with no result cached in the application. Ground truth will be shown only after prediction for benchmark examples.

- We will open the website's public page and submit a raw amino-acid sequence. We will demonstrate input validation with one intentionally invalid character.
- The inference worker will produce per-residue embeddings, domain assignments, and CATH labels.
- We will inspect the linear domain track. Selecting a domain will highlight its intervals, CATH hierarchy, and confidence values.
- We will open the similar-domain panel and trace one result to its CATH or Pfam record. We will explain that the displayed potential function comes from supporting database evidence.
- We will generate or load the optional predicted structure and show that the same domain colors appear in the three-dimensional viewer.
- For the annotated multidomain example, we will reveal the ground truth and compare Domaino with CHAINSAW using intersection over union, boundary error, and predicted domain count.
- We will conclude by presenting the full held-out benchmark results, including segmentation, classification, end-to-end performance, inference time, and the principal limitations of the final system.

2.4 Impact

Domaino will provide a lower-input-cost route to domain segmentation and classification. Sequence data is routinely available before an experimental or computed structure, so a sequence-only parser can support early exploration of newly identified and designed proteins. Researchers can use predicted boundaries to select constructs, compare domain architectures, prioritize regions for experiments, or decide whether a more expensive structural analysis is worthwhile. The integrated viewer reduces the effort required to connect a model output with CATH, Pfam, and three-dimensional context.

The project will produce a reusable data-obtaining pipeline, a structured inference API, a benchmarking pipeline, and deployment artifacts. These components can support later models even if one experimental architecture is replaced.

2.5 Risk Analysis

Risk 1: CATH, Pfam, and TED may give different boundaries for the same protein

Impact: High

Likelihood: High

Why and how the risk may occur: The three sources do not define domains in exactly the same way. CATH assigns structural domains using experimentally determined protein structures. TED derives domains from AlphaFold-predicted structures using multiple structural segmentation methods. Pfam identifies sequence regions belonging to conserved protein families, which may correspond to a complete structural domain, only part of one, or occasionally more than one structural unit. Consequently, the same residues may be labeled as a domain in one source and as a linker or different region in another. If these annotations are combined as though they were equivalent, the model may receive contradictory targets and learn uncertain boundaries.

Mitigation and how it reduces the risk: Every example will retain its source, database version, and available annotation types. CATH will serve as the principal ground truth for supervised segmentation, CATH classification, validation, and final testing. Pfam will provide auxiliary sequence-region supervision but will not be treated as a source of CATH labels. TED may be used for lower-weight pretraining or representation learning, followed by fine-tuning on experimental CATH data. Source-specific loss masks will ensure that a training example contributes only to targets actually provided by that source. Agreement between sources will also be measured before training.

Risk 2: Closely related proteins may appear in both training and evaluation data

Impact: High

Likelihood: High

Why and how the risk may occur: Randomly splitting individual protein sequences is insufficient because biological databases contain many homologous proteins inherited from a common ancestor. Two sequences can be different database entries while still having nearly identical domain organization and CATH labels. If one relative is used for training and another for testing, the model may appear to generalize when it is effectively recognizing a familiar family. Such homology leakage would inflate validation scores and make comparisons with CHAINSAW and CATHe2 unreliable.

Mitigation and how it reduces the risk: Sequences will first be grouped using CATH sequence clusters, such as S35 clusters, or equivalent MMseqs2 similarity clusters. Every member of a cluster will be assigned to only one of the training, validation, or test partitions. The released CHAINSAW benchmark chains and the CATHe2 test domains will be reserved before constructing the training set, and close homologues will also be removed from training. The split procedure, thresholds, removed overlaps, and final sequence counts will be documented. This makes the evaluation measure performance on genuinely unfamiliar proteins and allows the reported results to be compared meaningfully with the published benchmarks.

Risk 3: Rare CATH Homologous-superfamily classes may not contain enough examples for reliable learning

Impact: High

Likelihood: High

Why and how the risk may occur: CATH has thousands of Homologous-superfamily, or H-level, classes, but their frequencies are highly unequal. A small number of superfamilies contain many domains, while many others contain only a few experimentally characterized examples. A standard classifier can therefore minimize its total error by favoring common classes and may produce apparently high overall accuracy while performing poorly on rare but scientifically important domains. Some test proteins may also belong to superfamilies absent from training.

Mitigation and how it reduces the risk: The frequency of every CATH class will be calculated before defining the final classification label space. Class-balanced or focal loss will reduce the dominance of common classes, and the hierarchical relationship among the C, A, T, and H levels will be incorporated so that an example can still receive a meaningful higher-level classification when its exact H-level class is uncertain. Predictions below a calibrated confidence threshold will be reported as unknown or uncertain instead of being forced into an unsupported class. Accuracy will be accompanied by macro F1, per-level F1, per-class recall, and results grouped by class frequency. This prevents success on common classes from hiding failure on rare classes and makes the limits of the classifier visible.

Risk 4: Sequence-only segmentation may perform below structure-based CHAINSAW, particularly for multidomain or discontinuous proteins

Impact: High

Likelihood: Medium

Why and how the risk may occur: CHAINSAW receives residue-distance and secondary-structure information derived from a three-dimensional structure. Domain boundaries that are clear in three-dimensional space may be difficult to infer from sequence alone, especially when sequence similarity is low. The problem is harder for multidomain proteins, short domains, domains separated by long linker regions, and discontinuous domains whose residues occur in separated sequence intervals but come together after folding. A model optimized mainly on single-domain proteins could obtain a good average score while still failing on the cases for which domain segmentation is most useful.

Mitigation and how it reduces the risk: Training batches will deliberately include sufficient multidomain, discontinuous-domain, short-domain, and difficult low-similarity examples. In addition to residue-level boundary prediction, the model will learn pairwise same-domain relationships so that separated intervals can be assigned to one domain. Thresholds and clustering parameters will be selected using only the validation set. Performance will be reported separately for single-domain, multidomain, continuous, and discontinuous cases, and ablation experiments will determine which design components improve each category. The published CHAINSAW benchmark and metrics will be used unchanged. These measures focus development on the difficult cases and prevent strong single-domain results from concealing weak multidomain performance.

Risk 5: Pairwise residue representations may exceed available GPU memory

Impact: High

Likelihood: Medium

Why and how the risk may occur: A pairwise segmentation model represents relationships between every pair of residues. Its size therefore grows quadratically with protein length: increasing a sequence from 500 to 2,000 residues increases the number of residue pairs from 250,000 to 4 million. During training, intermediate feature maps, gradients, and multiple network layers require considerably more memory than the final matrix alone. A small number of very long proteins could consequently cause out-of-memory failures, severely reduce batch size, or make training impractically slow.

Mitigation and how it reduces the risk: Proteins will be grouped into length-based batches so that short sequences do not inherit the memory requirements of the longest sequence in a batch. Mixed-precision training, gradient accumulation, and pair subsampling will be used where they preserve accuracy. Long proteins will be handled using overlapping windows with reconciliation of predictions in shared regions. If the full pairwise representation remains impractical, sparse attention or a low-rank pair representation will be evaluated as a fallback. A maximum supported sequence length will be measured experimentally and displayed to users rather than being left implicit. These measures bound memory use while maintaining coverage of domain relationships across long sequences.

Risk 6: Optional three-dimensional structure prediction may be too slow for interactive use

Impact: Medium

Likelihood: Medium

Why and how the risk may occur: Domaino's sequence-based inference can be completed independently, but generating a predicted three-dimensional structure for visualization requires an additional model such as AlphaFold or ESMFold. Structure prediction may require substantial GPU memory and processing time, and delays may increase when several users submit jobs simultaneously. If the web interface waits for this stage

before showing any output, a slow or failed structure job could make the entire system appear unresponsive even when domain prediction succeeded.

Mitigation and how it reduces the risk: Domain segmentation and CATH classification will be completed and returned before optional structure generation begins. Structure prediction will run as a separate asynchronous job with visible states such as queued, running, completed, and failed. Previously generated structures will be cached when appropriate, and the final demonstration will include precomputed examples as a fallback. Timeouts or structure-prediction failures will not remove the sequence-based results. This separation ensures that the central Domaino function remains usable even when optional visualization is delayed or unavailable.

Risk 7: External databases or network services may be unavailable or may change during development

Impact: Medium

Likelihood: Medium

Why and how the risk may occur: Training, similarity search, functional annotation, and visualization may use data from resources such as CATH, Pfam, the Protein Data Bank, UniProt, or AlphaFold DB. Their websites or APIs can experience downtime, enforce request limits, change response formats, or publish a newer release with different identifiers. A live dependency could therefore interrupt a demonstration or make an experiment impossible to reproduce later.

Mitigation and how it reduces the risk: The exact releases used for training and benchmarking will be downloaded, checksummed, versioned, and stored locally. Essential domain labels and metadata required for core inference will be available from local files rather than requested from live services. Responses used for optional similarity and functional information will be cached, and failures will produce a clear “temporarily unavailable” message without affecting domain prediction. Recording release versions and preprocessing steps will allow the same experiments to be repeated even after an external database changes.

2.6 Gantt Chart

Figure 2 shows the planned fourteen-week schedule.

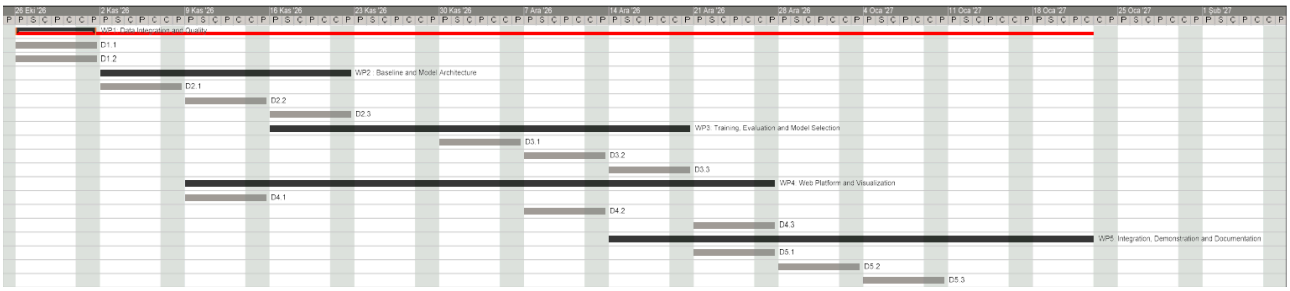


Figure 3: Gantt chart showing Work Packages and Deliverables

Section 3: Economical and Ethical Issues

3.1 Economic feasibility

Domaino is registered as a software-only project and does not require new laboratory equipment. CATH,

Pfam, PDB data, the core modeling libraries, and the proposed web stack are available publicly. The primary scarce resources are GPU time and storage. Training will use the university computing cluster and Slurm, which contributors have access to.

3.2 Engineering ethics

Our ethical duties are to avoid harm, follow the limitations, respect the rights and privacy, and make the system easy to check out. These goals are for professional ethics in engineering fields [19], [20]. Domain boundaries, CATH labels, similarity, and potential function are predictions. The interface will separate model confidence from database evidence and will not label an output experimentally verified unless its source record is experimental. PDB and CATH overrepresent proteins that are experimentally tractable and well-studied. Pfam and TED inherit additional taxonomic and annotation biases. Results will therefore be reported by class, length, and domain count, and low-coverage cases will receive an unknown or low-confidence result. Overlapping data and repeated protein chains can produce misleading performance results. To prevent this, data splits, exclusion lists, and test evaluation files will be version-controlled. We will not leave out negative or unclear comparison results. Newly designed protein sequences may be private intellectual property. The web app will save data for a short time, restrict access to saved jobs, allow users to delete their data, and never reuse submitted data to train models without clear consent. Domaino is a tool for labeling and visualizing proteins, not for creating them. However, information about protein function can be used for both helpful and harmful biological work. The public demo will limit batch processing, record data sources, and avoid claiming that the software checks for safety or danger. Databases, model weights, and visualization libraries (like CATH, Pfam, and PDB) have their own usage terms. The release will include a third-party notices file and proper database citations, and it will not share files beyond what the terms allow. To save energy and computing resources, the project will reuse pre-trained models, use efficient calculation methods, and stop training early to avoid unnecessary tests. For accessibility, domain colors will use a colorblind-friendly palette with text labels, and error messages will explain wrong inputs instead of failing silently.

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